

1. Find the equation of the tangent plane to the sphere $x^2 + y^2 + z^2 = 14$ at the point $(1, 2, 3)$.

2. Find the equation of the tangent plane at the point $(0, 9, 4)$ to the surface $z = e^x \sqrt{y} + x^2 + 1$.

3. Let $x^{10} + x^{10}y^{10} + x^{10}y^{20}z^{30} = 3$

a) Find the equation of the tangent plane at the point $(1,1,1)$ to the surface $x^{10} + x^{10}y^{10} + x^{10}y^{20}z^{30} = 3$.

b) Determine the linear approximation of the function $x^{10} + x^{10}y^{10} + x^{10}y^{20}z^{30} = 3$ at $(1,1,1)$ and use it to approximate the value of the z -coordinate of the surface at the point $(0.8,0.9)$.

4. Let $f(x, y) = 2y \sin(x - y)$.

a) Find an equation of the tangent plane to $f(x, y)$ at the point $(1, 1, 0)$.

b) Determine the linear approximation of the function $f(x, y)$ at $(1, 1)$ and use it to approximate $f(1.2, 0.9)$.